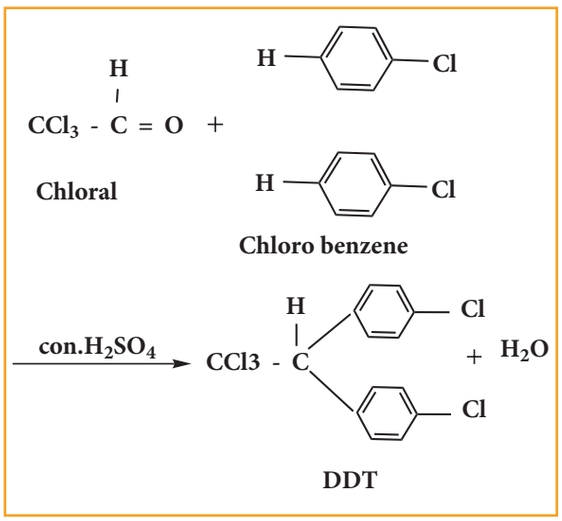
14.6.5 DDT (p,p'-dichloro diphenyl trichloro ethane)

DDT, the first chlorinated organic pesticide was prepared in 1873, and in 1939 Paul Muller discovered the effectiveness of DDT as an insecticide. He was awarded Noble prize in medicine and physiology in 1948 for this discovery.

DDT can be prepared by heating a mixture of chlorobenzene with chloral (Trichloro acetaldehyde) in the presence of Conc.H_2SO_4.



Evaluate Yourself

8) What is the IUPAC name of the insecticide DDT? Why is their use banned in most of the countries?

Uses:

i) DDT is used to control certain insects which carries diseases like malaria and yellow fever

ii) It is used in farms to control some agricultural pests

iii) It is used in building construction as pest control

iv) It is used to kill various insects like housefly and mosquitoes due to its high and specific toxicity.

SUMMARY

- The compounds obtained by the substitution of hydrogen atom of alkanes by halogen atom are called haloalkane, while the compounds obtained by the substitution of hydrogen atoms of arenes by halogen atom are called haloarenes.
- Modern classification of halo compounds is based on the halogen with carbon possessing sp^3 hybridisation. In these compounds the electronegativity of halogen is more than that of carbon, hence $C^{\delta+} - X^{\delta-}$ bond becomes polar.
- Haloalkane
- Haloalkanes are prepared from alkanes, alkenes or alcohols. The boiling points of haloalkane are higher than that of corresponding hydrocarbons.
- Haloalkane undergoes nucleophilic substitution and elimination reactions. Primary alkyl halides undergo S_N2 mechanism. If the reactant is chiral, the product formed exhibits inversion of stereo chemical configuration Tertiary alkyl halide undergoes S_N1 mechanism, via carbonium ion formation. If the reactant is chiral, the product formed is optically inactive due to racemisation.
- Organo metallic compound
- Haloalkane reacts with metal to form organometallic compounds like Grignard reagent. It is represented as $R^{\delta-}–Mg^{\delta+}X$. Grignard reagent reacts with variety of substances to give almost all class of organic compounds like alcohols, aldehydes, ketones, acids etc.
- Haloarenes

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